

# Convolutional Coded Poisson Receivers

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**Abstract**—In this paper, we present a framework for convolutional coded Poisson receivers (CCPRs) that incorporates spatially coupled methods into the architecture of coded Poisson receivers (CPRs). We use density evolution equations to track the packet decoding process with the successive interference cancellation (SIC) technique. We derive outer bounds for the stability region of CPRs when the underlying channel can be modeled by a  $\phi$ -ALOHA receiver. The stability region is the set of loads that every packet can be successfully received with a probability of 1. Our outer bounds extend those of the spatially-coupled Irregular Repetition Slotted ALOHA (IRSA) protocol and apply to channel models with multiple traffic classes. For CCPRs with a single class of users, the stability region is reduced to an interval. Therefore, it can be characterized by a percolation threshold. We study the potential threshold by the potential function of the base CPR used for constructing a CCPR. In addition, we prove that the CCPR is stable under a technical condition for the window size. For the multiclass scenario, we recursively evaluate the density evolution equations to determine the boundaries of the stability region. Numerical results demonstrate that the stability region of CCPRs can be enlarged compared to that of CPRs by leveraging the spatially-coupled method. Moreover, the stability region of CCPRs is close to our outer bounds when the window size is large.

**Index Terms**—Coded Poisson receivers, irregular repetition slotted ALOHA, density evolution, potential function, successive interference cancellation.

## I. INTRODUCTION

THE fifth-generation networks (5G) and beyond aim to support various connectivity classes of users that have different quality-of-service (QoS) requirements, including (i) enhanced mobile broadband (eMBB), (ii) ultra-reliable low-latency communications (URLLC), and (iii) massive machine-type communications (mMTC) (see, e.g., [3], [4], [5], [6] and References therein). *Network slicing* [7] that partitions and allocates available radio resources to provide differentiated QoS in a *single* radio network has received much attention lately. For downlink transmissions, network slicing could be tackled by centralized scheduling algorithms [8].

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On the other hand, various schemes of coded multiple access (see, e.g., [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23]) have been proposed in the literature to address the problem of uplink transmissions. The framework of *Poisson receivers* [24] is an abstraction of these coded multiple access schemes to hide the encoding/decoding complexity of the physical layer from the Medium Access Control (MAC) layer.

A Poisson receiver specifies the probability that a packet is successfully received (decoded) when the number of packets transmitted simultaneously to the receiver follows a Poisson distribution (Poisson offered load). Inspired by Irregular Repetition Slotted ALOHA (IRSA) [10] and coded slotted ALOHA (CSA) [11], [12], [13], [14], [15], [16], a Poisson receiver can be used as a building block to construct a system of coded Poisson receivers (CPR) with multiple classes of users and multiple classes of receivers. In [25], the stability region of a system of CPR is defined as the region of Poisson offered loads in which all the packets can be successfully received (with probability 1). The stability regions of various network slicing policies, including the complete sharing and the receiver reservation policies, were computed numerically in [25]. The numerical results in [25] showed that different network slicing policies lead to different stability regions. Given a targeted stability region in a network with various classes of traffic, one engineering problem is to design a system of CPR associated with a network slicing policy to cover the targeted stability region.

One objective of this paper is to explore the boundaries of the stability region of a system of CPRs and to devise schemes to approach the boundaries. The contributions of the paper are as follows:

(i) *Outer bounds for the stability region* (see Theorem 4): For IRSA and CSA, there is only a single class of users and a single class of receivers. Thus, the stability region is reduced to an interval and its boundary is limited to one packet per time slot (as the underlying channel is the slotted ALOHA channel). When the underlying channel is replaced by the  $D$ -fold ALOHA channel (in which, at most,  $D$  packets can be successfully received in a time slot), the boundary is limited to  $D$  packets per time slot. These boundaries are simply the capacities of the underlying channels. As an application of the Area Theorem [26], tighter bounds for the stability regions for IRSA and coded  $D$ -fold ALOHA were derived in [13] and [27], respectively. In Theorem 4, we derive outer bounds for the stability region of a system of CPR when the underlying channel is modeled by a  $\phi$ -ALOHA receiver [28]. Such a model is a generalization of both the slotted ALOHA channel and the  $D$ -fold ALOHA channel. As such, our outer bounds

recover the upper bounds for IRSA and coded  $D$ -fold ALOHA in [13] and [27] as special cases. Our outer bounds can also be applied to the setting with multiple traffic classes.

(ii) *Threshold saturation of Convolutional Coded Poisson Receivers* (see Theorem 5): Motivated by the remarkable performance of convolutional Low-Density Parity-Check (LDPC) codes (see, e.g., [29], [30], [31]) and the spatially coupled IRSA [27], [32], we propose using the construction method for convolutional LDPC codes and spatially coupled IRSA to construct Convolutional Coded Poisson Receivers (CCPRs) with multiple classes of users and multiple classes of receivers. To construct a CCPR, one first constructs  $L$  stages of random bipartite graphs from a base CPR. The  $L$  stages of bipartite graphs are concatenated to form a single bipartite graph. Then, we rewire the receiver ends of the  $\ell$ -th copy, where  $1 \leq \ell \leq L$ , to a randomly selected copy in a window  $w$ . We refer the reader to Section IV-A for more details on the constructions. By using the *shortening* technique in LDPC codes, we show in Theorem 5 that the stability region with  $L$  stages is not larger than that with  $L-1$  stages, and thus the stability region converges as  $L \rightarrow \infty$ . Such a phenomenon is analogous to *threshold saturation* for convolutional LDPC codes and spatially coupled IRSA (see, e.g., [29], [31], [32], [33]).

(iii) *Potential functions and percolation thresholds for CCPRs with a single class of users* (see Theorem 6 and Theorem 7): Even though we know that the stability region converges as  $L \rightarrow \infty$ , characterizing such a stability region is difficult for CCPR with multiple classes of users. For CCPRs with a single class of users, we use a remarkable mathematical tool called *potential functions* by Yedla et al. [30], [34], [35] in their study of LDPC codes. In Theorem 6, we derive the potential function of a (base) CPR. Based on such a potential function, we define three thresholds: the single-system threshold  $G_s^*$ , the potential threshold  $G_{conv}^*$ , and the potential bound  $G_{up}^*$ . We show the threshold saturation theorem in Theorem 7 that for all  $G \leq G_{conv}^*$ , a CCPR with  $L$  stages is stable if the window size  $w$  satisfies a technical condition. The three thresholds are related as follows:

$$G_s^* < G_{conv}^* < G_{up}^*. \quad (1)$$

The single-system threshold  $G_s^*$  recovers the percolation threshold of a (base) CPR in [25]. When the underlying channel is modeled by the  $D$ -fold ALOHA, the potential bound  $G_{up}^*$  is the same as the upper bound for stability in [27].

(iv) *Numerical results* (see Section VI): We provide numerical results for various systems of convolutional coded Poisson receivers. For CCPRs with one class of users and one class of receivers, our numerical results for  $G_{conv}^*$  and  $G_{up}^*$  in various parameter settings match very well with those in Table I of [32] and Table I of [27]. We also study the convolutional IRSA with two classes of users and two classes of receivers. We consider the two network slicing policies in [25]: the complete sharing policy and the receiver reservation policy. For  $L = 40$ , our numerical results show that the convolutional effect (due to

spatial coupling) can indeed enlarge the stability region. Such enlargement appears to be monotone in the window size  $w$ .

The structure of this paper is organized as follows: In Section II, we provide a review of the coded Poisson receiver framework. In Section III, we show the outer bounds of the stability region of CPRs. In Section IV, we introduce the concept of CCPRs and present the development of density evolution equations for their decoding process, as well as the threshold saturation phenomena. In Section V, we delve into the application of the potential function to our CCPR framework, focusing on a single class of users. We utilize potential functions to characterize three significant thresholds. We establish the saturation theorem in this section as well. Some numerical results for both single and multiple traffic classes are presented in Section VI. We conclude the paper and discuss potential extensions of our work in Section VII.

## II. REVIEW OF THE FRAMEWORK OF CODED POISSON RECEIVERS

For the paper to be self-contained, we briefly review the framework of coded Poisson receivers in [24], [25], and [28].

### A. Poisson Receivers and ALOHA Receivers

A Poisson receiver is an *abstract* receiver proposed in [24]. The key insight of a Poisson receiver is to specify the packet success probability when the system is subject to Poisson arrivals. Specifically, a system with  $K$  classes of input traffic is said to have a Poisson offered load  $\rho = (\rho_1, \rho_2, \dots, \rho_K) \in \mathbb{R}_{\geq 0}^K$  if these  $K$  classes of input traffic are *independent*, and the number of class  $k$  packets arriving at the system follows a Poisson distribution with mean  $\rho_k$ , for  $k = 1, 2, \dots, K$ .

*Definition 1 (Poisson Receiver With Multiple Classes of Input Traffic [24]):* An abstract receiver is called a  $(P_{\text{suc},1}(\rho), P_{\text{suc},2}(\rho), \dots, P_{\text{suc},K}(\rho))$ -Poisson receiver with  $K$  classes of input traffic if the receiver is subject to a Poisson offered load  $\rho = (\rho_1, \rho_2, \dots, \rho_K) \in \mathbb{R}_{\geq 0}^K$ , a tagged (randomly selected) class  $k$  packet is successfully received with probability  $P_{\text{suc},k}(\rho)$ , for  $k = 1, 2, \dots, K$ .

The throughput of class  $k$  packets (defined as the expected number of class  $k$  packets that are successfully received) for a  $(P_{\text{suc},1}(\rho), P_{\text{suc},2}(\rho), \dots, P_{\text{suc},K}(\rho))$ -Poisson receiver subject to a Poisson offered load  $\rho$  is thus

$$\Theta_k = \rho_k \cdot P_{\text{suc},k}(\rho), \quad (2)$$

$k = 1, 2, \dots, K$ .

By viewing each time slot as a Poisson receiver, various well-known systems can be modeled by Poisson receivers in [24], [25], and [28]. These include Slotted ALOHA (SA) [36], SA with multiple cooperative receivers, and Rayleigh block fading channel with capture [16], [37], [38]. It can also be applied to the setting with multiple packet reception in a time slot (see, e.g., [21], [22], [27], [39], [40], [41]).

Like a Poisson receiver, an ALOHA receiver [28] is also an abstract receiver. Such an abstract receiver treats the physical layer with a *deterministic* input-output function. Denote by  $\mathbb{Z}^+$  the set of nonnegative integers. We say a system with

$K$  classes of input traffic is subject to a *deterministic* load  $n = (n_1, n_2, \dots, n_K) \in \mathcal{Z}^{+K}$  if the number of class  $k$  packets arriving at the system is  $n_k$ .

**Definition 2 (ALOHA Receiver With Multiple Classes of Input Traffic [28]):** Consider a deterministic function

$$\phi : \mathcal{Z}^{+K} \rightarrow \mathcal{Z}^{+K}$$

that maps a  $K$ -vector  $n = (n_1, n_2, \dots, n_K)$  to the  $K$ -vector  $(\phi_1(n), \phi_2(n), \dots, \phi_K(n))$ . An abstract receiver is called a  $\phi$ -ALOHA receiver (with  $K$  classes of input traffic) if the number of class  $k$  packets that are successfully received is exactly  $\phi_k(n)$ ,  $k = 1, 2, \dots, K$ , when the receiver is subject to a deterministic load  $n = (n_1, n_2, \dots, n_K)$ .

One typical example of the  $\phi$ -ALOHA receiver is a time slot in the slotted ALOHA (SA) system, where at most one packet can be received. The  $D$ -fold ALOHA system proposed in [21] is a generalization of the SA system. If there are less than or equal to  $D$  packets transmitted in a time slot, then all these packets can be successfully received. On the other hand, if there are more than  $D$  packets transmitted in a time slot, then all these packets are lost. Thus, a time slot in the  $D$ -fold ALOHA system is a  $\phi$ -ALOHA receiver (with a single class of input traffic), where

$$\phi(n) = \begin{cases} n & \text{if } n \leq D \\ 0 & \text{otherwise.} \end{cases} \quad (3)$$

Another example of a  $\phi$ -ALOHA receiver is the model for the near-far [21] decoding scheme. In such a model, there are two classes of input traffic and the power from one class (the near users) is much stronger than that of the other class (the far users). By decoding the near users first and using the SIC technique, a time slot in the near-far SIC decoding scheme can be modeled by a  $\phi$ -ALOHA with

$$\phi(n_1, n_2) = \begin{cases} (n_1, n_2) & \text{if } (n_1, n_2) \leq (1, 1) \\ (0, 0) & \text{otherwise.} \end{cases} \quad (4)$$

It was shown in Theorem 14 of [28] that for every ALOHA receiver, there is an induced Poisson receiver. This is done by computing the throughput of the  $\phi$ -ALOHA receiver when it is subject to the Poisson offered load  $\rho$  and then using (2) to find the success probability function. For the  $D$ -fold ALOHA system, the throughput for a Poisson offered load  $\rho$  is

$$\sum_{t=1}^D t \frac{e^{-\rho} \rho^t}{t!} = \rho \sum_{t=0}^{D-1} \frac{e^{-\rho} \rho^t}{t!}.$$

From (2), it is a Poisson receiver with the following success probability function:

$$P_{\text{suc}}(\rho) = \sum_{t=0}^{D-1} \frac{e^{-\rho} \rho^t}{t!}. \quad (5)$$

### B. Coded Poisson Receivers With Multiple Classes of Users and Receivers

The idea of using the SIC technique in the Contention Resolution Diversity Slotted ALOHA (CRDSA) protocol [9]

and the Irregular Repetition Slotted ALOHA (IRSA) protocol leads to the development of coded Poisson receivers (CPR) with multiple classes of users and receivers in [25]. As in [25], let us consider a system with  $G_k T$  class  $k$  users,  $k = 1, 2, \dots, K$ , and  $F_j T$  class  $j$  Poisson receivers,  $j = 1, 2, \dots, J$ , where  $T$  is a scaling parameter. In IRSA,  $T$  is the number of time slots. For communication systems with multiple orthogonal channels,  $T$  can be interpreted as the product of the number of channels and the number of time slots. Class  $j$  Poisson receivers have the success probability functions  $P_{\text{suc},1,j}(\rho), P_{\text{suc},2,j}(\rho), \dots, P_{\text{suc},K,j}(\rho)$  for the  $K$  classes of input traffic. Each class  $k$  user transmits its packet for (a random number of)  $L_k \geq 1$  times (copies). With the routing probability  $r_{k,j}$  ( $\sum_{j=1}^J r_{k,j} = 1$ ), each copy of a class  $k$  packet is transmitted *uniformly* and *independently* to one of the  $F_j T$  class  $j$  Poisson receivers.

There are two assumptions made in [25]:

- (i) Perfect SIC: as long as one copy of a packet is successfully received by one of the receivers, then it can be used to remove the other copies of that packet from the other receivers.
- (ii) Independent Poisson receivers: the event that a packet is successfully received by a Poisson receiver is independent of the outcomes of the other Poisson receivers as long as their input traffic is independent of each other.

As in [10] and [25], let  $\Lambda_{k,d}$  be the probability that a class  $k$  packet is transmitted  $d$  times, i.e.,

$$P(L_k = d) = \Lambda_{k,d}, \quad d = 1, 2, \dots \quad (6)$$

Define the generating function

$$\Lambda_k(x) = \sum_{d=0}^{\infty} \Lambda_{k,d} \cdot x^d \quad (7)$$

of the *degree distribution* of a class  $k$  user node, and the generating function

$$\lambda_k(x) = \sum_{d=0}^{\infty} \lambda_{k,d} \cdot x^d \quad (8)$$

of the *excess degree distribution* of a class  $k$  user node, where

$$\lambda_{k,d} = \frac{\Lambda_{k,d+1} \cdot (d+1)}{\sum_{d=0}^{\infty} \Lambda_{k,d+1} \cdot (d+1)} \quad (9)$$

is the probability that the user end of a randomly selected class  $k$  edge has additional  $d$  edges, excluding the randomly selected class  $k$  edge. Note that the mean degree of a class  $k$  user node is

$$\Lambda'_k(1) = \sum_{d=0}^{\infty} d \cdot \Lambda_{k,d}, \quad (10)$$

and that

$$\lambda_k(x) = \frac{\Lambda'_k(x)}{\Lambda'_k(1)}. \quad (11)$$

Let

$$G = (G_1, G_2, \dots, G_K), \\ \Lambda'(x) = (\Lambda'_1(x), \Lambda'_2(x), \dots, \Lambda'_K(x)),$$

$$R_j = \left( \frac{r_{1,j}}{F_j}, \frac{r_{2,j}}{F_j}, \dots, \frac{r_{K,j}}{F_j} \right), \quad (12)$$

and

$$\tilde{\rho}_j = G \circ \Lambda'(1) \circ R_j, \quad (13)$$

where  $\circ$  denotes the element-wise multiplication of two vectors.

Using the tree evaluation (density evolution) method in [10], [12], [42], and [43] and the reduced Poisson offered load argument in [24], the following result was derived in Theorem 1 of [24].

**Theorem 1 (Theorem 1 of [24]):** As  $T \rightarrow \infty$ , the system of CPRs after the  $i$ -th SIC iteration converges to a Poisson receiver with the success probability function for the class  $k$  traffic

$$\tilde{P}_{\text{suc},k}^{(i)}(G) = 1 - \Lambda_k \left( 1 - \sum_{j=1}^J r_{k,j} P_{\text{suc},k,j} \left( q^{(i-1)} \circ \tilde{\rho}_j \right) \right), \quad (14)$$

$k = 1, 2, \dots, K$ , where  $q^{(i)} = (q_1^{(i)}, q_2^{(i)}, \dots, q_K^{(i)})$  can be computed recursively from the following equation:

$$q_k^{(i+1)} = \lambda_k \left( 1 - \sum_{j=1}^J r_{k,j} P_{\text{suc},k,j} \left( q^{(i)} \circ \tilde{\rho}_j \right) \right), \quad (15)$$

with  $q^{(0)} = (1, 1, \dots, 1)$ .

If the number of copies  $L_k \geq 2$  for all  $k$ , i.e.,  $\Lambda_{k,1} = 0$ , then the generating function  $\lambda_k(x)$  is strictly increasing for  $0 \leq x \leq 1$ , and thus invertible. By letting  $p_k^{(i+1)} = \lambda_k^{-1}(q_k^{(i+1)})$ , one can rewrite (15) as follows:

$$p_k^{(i+1)} = 1 - \sum_{j=1}^J r_{k,j} P_{\text{suc},k,j} (G \circ \Lambda'(p^{(i)}) \circ R_j), \quad (16)$$

where  $p^{(i)} = (p_1^{(i)}, p_2^{(i)}, \dots, p_K^{(i)})$  and  $p^{(0)} = (1, 1, \dots, 1)$ .

Note that a system of CPRs with  $K$  classes of users and  $J$  classes of receivers is characterized by the following parameters: (i) the total number of Poisson receivers  $T$ , (ii) the offered load vector  $G = (G_1, G_2, \dots, G_K)$ , (iii) the degree distribution vector  $\Lambda(x) = (\Lambda_1(x), \Lambda_2(x), \dots, \Lambda_K(x))$ , (iv) the  $K \times J$  routing matrix  $R = (r_{k,j})$ , and (v) the partition vector  $F = (F_1, F_2, \dots, F_J)$ . In [25], the routing matrix and the partition vector correspond to network slicing policies. To ease the presentation, we denote a system of CPRs with  $K$  classes of users and  $J$  classes of receivers as the  $(T, G, \Lambda(x), R, F)$ -CPR.

### C. Stability

The stability of a system of coded Poisson receivers in [25] is motivated by the stability of a queueing system, where the system is (rate) stable if the departure rate is the same as the arrival rate. In that regard, a system of coded Poisson receivers is stable if all its packets can be successfully received. The precise definition is given below.

**Definition 3 (Stability of Coded Poisson Receivers With Multiple Classes of Input Traffic [25]):** Consider the  $(T, G, \Lambda(x), R, F)$ -CPR described in Section II-B. A Poisson

offered load  $G = (G_1, G_2, \dots, G_K) \in \mathbb{R}_{\geq 0}^K$  to a system of coded Poisson receivers is said to be *stable* if, as  $T \rightarrow \infty$ , the probability that a packet is successfully received approaches 1 when the number of iterations goes to infinity, i.e.,

$$\lim_{i \rightarrow \infty} \tilde{P}_{\text{suc},k}^{(i)}(G) = 1, \quad k = 1, 2, \dots, K, \quad (17)$$

where  $\tilde{P}_{\text{suc},k}^{(i)}(G)$  is defined in (14).

As in [25], we make the following four assumptions for the stability analysis:

- (A1) For all  $k = 1, \dots, K$  and  $j = 1, 2, \dots, J$ , the success probability function  $P_{\text{suc},k,j}(\rho)$  is a continuous and decreasing function of  $\rho$ , and  $P_{\text{suc},k,j}(\mathbf{0}) = 1$ , where  $\mathbf{0}$  is the vector with all its elements being 0. Furthermore,  $P_{\text{suc},k,j}(\rho)$  is assumed to have a continuous second derivative.
- (A2) If  $\rho \neq \mathbf{0}$ , then  $P_{\text{suc},k,j}(\rho) < 1$  for all  $k = 1, \dots, K$ , and  $j = 1, 2, \dots, J$ .
- (A3)  $r_{k,j} > 0$  for all  $k = 1, \dots, K$ , and  $j = 1, 2, \dots, J$ .
- (A4) Every packet is transmitted at least twice, i.e.,  $\Lambda_{k,1} = 0$  for all  $k = 1, 2, \dots, K$ .

Under these assumptions, a necessary and sufficient condition for a Poisson offered load  $G$  to be stable is presented in Theorem 2. A monotonicity result is presented in Theorem 3. These two theorems appear in Theorems 2 and 3 in [25]. They are duplicated here for their importance and for the completeness of this paper.

**Theorem 2 (Theorem 2 of [25]):** Under (A1), a Poisson offered load  $G \in \mathbb{R}_{\geq 0}^K$  is stable if  $q = \mathbf{0}$  is the unique solution in  $[0, 1]^K$  of the following  $K$  equations:

$$q_k = \lambda_k \left( 1 - \sum_{j=1}^J r_{k,j} P_{\text{suc},k,j} (q \circ \tilde{\rho}_j) \right), \quad (18)$$

$k = 1, 2, \dots, K$ , where  $\tilde{\rho}_j$  is defined in (13) and  $q = (q_1, q_2, \dots, q_K)$ . On the other hand, under (A1), (A2), and (A3), a positive Poisson offered load  $G$  (with  $G_k > 0$  for all  $k$ ) is stable only if  $q = \mathbf{0}$  is the unique solution in  $[0, 1]^K$  of the  $K$  equations in (18). Moreover, by (16), under (A1), (A2), (A3) and (A4), a positive Poisson offered load  $G$  (with  $G_k > 0$  for all  $k$ ) is stable if and only if  $p = \mathbf{0}$  is the unique solution in  $[0, 1]^K$  of the following  $K$  equations:

$$p_k = 1 - \sum_{j=1}^J r_{k,j} P_{\text{suc},k,j} (G \circ \Lambda'(p) \circ R_j), \quad (19)$$

$k = 1, 2, \dots, K$  and  $p = (p_1, p_2, \dots, p_K)$ .

**Theorem 3 (Theorem 3 in [25]):** Suppose that (A1), (A2), and (A3) hold. If a positive Poisson offered load  $\hat{G} = (\hat{G}_1, \hat{G}_2, \dots, \hat{G}_K) \in \mathbb{R}_{\geq 0}^K$  is stable, then any Poisson offered load  $G$  with  $G \leq \hat{G}$  is also stable.

The monotonicity results in Theorem 3 leads to the notion of the stability region in [25].

**Definition 4:** Under (A1), (A2), and (A3), the stability region  $S$  is defined as the maximal stable set such that (i) any  $G \in S$  is stable, and (ii) any  $G \notin S$  is not stable.



### III. OUTER BOUNDS FOR THE STABILITY REGION

In this section, we derive an outer bound for the stability region for a system of coded Poisson receivers when the Poisson receivers are induced from  $\phi$ -ALOHA receivers. For this, we need to define the concept of “capacity” for  $\phi$ -ALOHA receivers. Analogous to the terminology used in queueing theory, we use “capacity” for systems with deterministic loads and “stability” for systems with stochastic loads.

For a  $\phi$ -ALOHA receiver, the *failure* function  $\phi^c$  [28] is defined by

$$\phi^c(n) = n - \phi(n). \quad (20)$$

A  $\phi$ -ALOHA receiver is called *monotone* if the *failure* function  $\phi^c$  is *increasing* in the deterministic load  $n$ , i.e., for any  $n' \leq n''$ ,

$$\phi^c(n') \leq \phi^c(n''). \quad (21)$$

The failure function represents the number of packets remaining to be decoded. With this definition of monotonicity, we now define the capacity region.

**Definition 5 (Capacity of a  $\phi$ -ALOHA Receiver):** For a monotone  $\phi$ -ALOHA receiver, its capacity region  $S$  is defined to be the set of deterministic loads such that all the packets are successfully received, i.e.,

$$S = \{n : \phi(n) = n\}. \quad (22)$$

An  $K$ -vector  $(b_1, b_2, \dots, b_K)$  is called an *affine capacity envelope* with the bound  $B$  if for  $n \in S$ ,

$$\sum_{k=1}^K b_k \phi_k(n) \leq B. \quad (23)$$

For instance, for the  $D$ -fold ALOHA system with  $K$  classes of users, it has the capacity region  $S = \{n : \sum_{k=1}^K n_k \leq D\}$  as the total number of packets that can be successfully received in a time slot is at most  $D$ . As such, the  $K$ -vector  $(1, 1, \dots, 1)$  is also an affine capacity envelope with the bound  $D$  for the  $D$ -fold ALOHA system with  $K$  classes of users.

We will use affine capacity envelopes to derive outer bounds for the stability region for a system of CPRs. For this, we need two properties of  $\phi$ -ALOHA receivers in [28]: the closure property and the all-or-nothing property. As the failure function represents the number of packets that remain to be decoded, we can decode the remaining packets for the second time by removing those successfully decoded packets. This corresponds to the SIC technique in the literature. The number of packets remaining to be decoded after the second time of decoding is  $\phi^c(\phi^c(n))$ . Intuitively, we can carry out the iterative decoding approach (for an infinite number of times) until no more packets can be decoded. A  $\phi$ -ALOHA receiver that always does the iterative decoding approach until no more packets can be decoded is said to satisfy the *closure* property, i.e.,

$$\phi^c(\phi^c(n)) = \phi^c(n), \quad (24)$$

for all  $n$ . In addition to this, we also need the all-or-nothing property. A  $\phi$ -ALOHA receiver with  $K$  classes of input traffic is said to be an *all-or-nothing* receiver if it is *monotone* and satisfies the *all-or-nothing property*, i.e., either  $\phi_k(n) = n_k$  or

$\phi_k(n) = 0$  for all  $n = (n_1, n_2, \dots, n_K)$  and  $k = 1, 2, \dots, K$ . The all-or-nothing property implies if class  $k$  packets (for some  $k$ ) are successfully decoded, then they are all decoded during the same iteration.

In the following theorem, we derive an outer bound for the stability region of a system of coded Poisson receivers by using the concept of the affine capacity envelope.

**Theorem 4:** Consider the  $(T, G, \Lambda(x), R, F)$ -CPR with  $T$  Poisson receivers being induced from a  $\phi$ -ALOHA receiver. Suppose that the  $\phi$ -ALOHA receiver that satisfies the closure property and the all-or-nothing property, and it has an affine capacity envelope  $(b_1, b_2, \dots, b_K)$  with the bound  $B$  and that  $b_k$ 's are binary. Then for any stable offered load  $G$ ,

$$\sum_{k=1}^K b_k G_k \leq \sum_{j=1}^J F_j \left( \sum_{\tau=0}^{B-1} \tau e^{-\mu_j} \frac{\mu_j^\tau}{\tau!} + B \left( 1 - \sum_{\tau=0}^{B-1} e^{-\mu_j} \frac{\mu_j^\tau}{\tau!} \right) \right), \quad (25)$$

where

$$\mu_j = \sum_{k=1}^K b_k G_k \Lambda'_k(1) r_{k,j} / F_j. \quad (26)$$

For the proof of Theorem 4, we need the following lemma.

**Lemma 1:** Under the assumptions in Theorem 4, the number of packets that are *actually* decoded by any one of the  $T$  Poisson receivers is within the capacity region of the  $\phi$ -ALOHA receiver.

**Proof:** Consider a particular Poisson receiver. During the SIC decoding process, we assume that there are  $n_{k_\ell} > 0$  class  $k_\ell$  packets that are decoded during the  $i_\ell$ -th iteration,  $\ell = 1, 2, \dots, M$  for some  $M$ . Without loss of generality, we assume that  $i_1 \leq i_2 \leq \dots \leq i_M$ . Let  $\mathbf{e}_k$  be the  $1 \times K$  vector with its  $k$ -th element being 1 and 0 otherwise. Thus, the number of packets decoded by this receiver can be represented by the vector  $\sum_{\ell=1}^M n_{k_\ell} \mathbf{e}_{k_\ell}$ . To prove this lemma, we need to show that the decoded packets by this Poisson receiver is within the capacity region of the  $\phi$ -ALOHA receiver, i.e.,

$$\phi \left( \sum_{\ell=1}^M n_{k_\ell} \mathbf{e}_{k_\ell} \right) = \sum_{\ell=1}^M n_{k_\ell} \mathbf{e}_{k_\ell} \quad (27)$$

This is equivalent to showing that

$$\phi_{k_\ell}^c \left( \sum_{\ell=1}^M n_{k_\ell} \mathbf{e}_{k_\ell} \right) = 0, \quad (28)$$

for  $\ell = 1, 2, \dots, M$ . From the all-or-nothing property, we know that  $M \leq K$  as packets of the same class are decoded in the same iteration.

We prove (28) by induction. Suppose that there are  $n^{(i)} = (n_1^{(i)}, n_2^{(i)}, \dots, n_K^{(i)})$  remaining packets at the receiver during the  $i$ -th iteration. As the number of decoded class  $k$  packets cannot be larger than the number of remaining class  $k$  packets, we have

$$n_{k_1} \mathbf{e}_{k_1} \leq n^{(i_1)}. \quad (29)$$

As there are  $n_{k_1} > 0$  class  $k_1$  packets that are decoded during the  $i_1$ -th iteration, we have from the monotone property and the all-or-nothing property that

$$\phi_{k_1}^c \left( \sum_{\ell=1}^M n_{k_\ell} \mathbf{e}_{k_\ell} \right) \leq \phi_{k_1}^c(n^{(i_1)}) = 0. \quad (30)$$

Now assume that (28) holds for  $\ell = 1, 2, \dots, m$  as the induction hypothesis. From the induction hypothesis, we have

$$\phi^c \left( \sum_{\ell=1}^M n_{k\ell} \mathbf{e}_{k\ell} \right) \leq \sum_{\ell=m+1}^M n_{k\ell} \mathbf{e}_{k\ell}. \quad (31)$$

From the closure property, the monotone property and (31), we know that

$$\begin{aligned} \phi_{k_{m+1}}^c \left( \sum_{\ell=1}^M n_{k\ell} \mathbf{e}_{k\ell} \right) &= \phi_{k_{m+1}}^c \left( \phi^c \left( \sum_{\ell=1}^M n_{k\ell} \mathbf{e}_{k\ell} \right) \right) \\ &\leq \phi_{k_{m+1}}^c \left( \sum_{\ell=m+1}^M n_{k\ell} \mathbf{e}_{k\ell} \right). \end{aligned} \quad (32)$$

As the number of decoded class  $k$  packets cannot be larger than the number of remaining class  $k$  packets, we have at the  $i_{m+1}$ -th iteration that

$$\sum_{\ell=m+1}^M n_{k\ell} \mathbf{e}_{k\ell} \leq n^{(i_{m+1})}. \quad (33)$$

It follows from the monotone property and the all-or-nothing property that

$$\phi_{k_{m+1}}^c \left( \sum_{\ell=m+1}^M n_{k\ell} \mathbf{e}_{k\ell} \right) \leq \phi_{k_{m+1}}^c (n^{(i_{m+1})}) = 0. \quad (34)$$

Thus,  $\phi_{k_{m+1}}^c \left( \sum_{\ell=1}^M n_{k\ell} \mathbf{e}_{k\ell} \right) = 0$ , and we complete the induction. ■

*Proof (Theorem 4):* Let  $X_k(t)$  be the number of class  $k$  packets sent to the  $t$ -th Poisson receiver and  $Y_k(t)$  be the number of class  $k$  packets that are *actually* decoded by the  $t$ -th Poisson receiver. In view of Lemma 1 and the definition of the affine capacity envelope in (23), we have

$$\sum_{k=1}^K b_k Y_k(t) \leq B. \quad (35)$$

Note that (35) holds trivially when nothing is decoded in the  $t$ -th Poisson receiver, i.e.,  $Y_k(t) = 0$  for all  $k$ . Since  $Y_k(t) \leq X_k(t)$ , we have from (35) that

$$\sum_{k=1}^K b_k Y_k(t) \leq \min \left[ \sum_{k=1}^K b_k X_k(t), B \right]. \quad (36)$$

Taking expectations on both sides of (36) yields

$$\sum_{k=1}^K b_k \mathbb{E}[Y_k(t)] \leq \mathbb{E} \left[ \min \left[ \sum_{k=1}^K b_k X_k(t), B \right] \right]. \quad (37)$$

If the  $t$ -th Poisson receiver is a class  $j$  receiver, then we have from (13) that  $X_k(t)$ ,  $k = 1, 2, \dots, K$ , are independent Poisson random variables with mean

$$\rho_{k,j} = G_k \Lambda'_k(1) r_{k,j} / F_j. \quad (38)$$

Since  $b'_k$ s are binary and the sum of independent Poisson random variables is still a Poisson random variable, we

know that  $\sum_{k=1}^K b_k X_k(t)$  is a Poisson random variable with mean  $\mu_j$  as

$$\sum_{k=1}^K b_k \rho_{k,j} = \sum_{k=1}^K b_k G_k \Lambda'_k(1) r_{k,j} / F_j = \mu_j. \quad (39)$$

As there are  $F_j T$  class  $j$  Poisson receivers, the probability that the  $t$ -th Poisson receiver is a class  $j$  Poisson receiver is  $F_j$ . Thus, the expectation in the right-hand side of (37) can be computed by using the Poisson distribution, and we have

$$\begin{aligned} \sum_{k=1}^K b_k \mathbb{E}[Y_k(t)] &\leq \sum_{j=1}^J F_j \left( \sum_{\tau=1}^{B-1} \tau e^{-\mu_j} \frac{\mu_j^\tau}{\tau!} + B \left( 1 - \sum_{\tau=1}^{B-1} e^{-\mu_j} \frac{\mu_j^\tau}{\tau!} \right) \right). \end{aligned} \quad (40)$$

For a system of CPRs to be stable (the departure rate must be the same as the arrival rate), we must have  $G_k = \mathbb{E}[Y_k(t)]$ . Using this in (40) leads to the upper bound for the stable region in (25). ■

*Example 1 (Spatially Coupled IRSA):* For the spatially coupled IRSA with  $D$ -multipacket reception capability in [27], every time slot is a  $D$ -fold ALOHA and thus has the affine capacity envelope  $(1, 1, \dots, 1)$  with the bound  $D$ . Since there is only one class of users ( $K = 1$ ) and one class of receivers ( $J = 1$ ) in [27], the upper bound in (25) can be further simplified as follows:

$$\begin{aligned} G &\leq \sum_{\tau=0}^{D-1} \tau e^{-G \Lambda'(1)} \frac{G \Lambda'(1)^\tau}{\tau!} \\ &\quad + D \left( 1 - \sum_{\tau=0}^{D-1} e^{-G \Lambda'(1)} \frac{G \Lambda'(1)^\tau}{\tau!} \right) \\ &= D - \sum_{\tau=0}^{D-1} (D - \tau) e^{-G \Lambda'(1)} \frac{G \Lambda'(1)^\tau}{\tau!}. \end{aligned} \quad (41)$$

This recovers the result in Theorem 1 of [27].

One straightforward extension of the upper bound in (41) is to consider a mixture of  $D$ -fold ALOHA receivers in [28]. Specifically, with probability  $\pi_D$  (that satisfies  $\sum_{D=1}^{D_{\max}} \pi_D = 1$  for some positive integer  $D_{\max}$ ), the (induced) Poisson receiver is selected from a  $D$ -fold ALOHA. For such a Poisson receiver, the success probability function (cf. (5)) is

$$P_{\text{suc}}(\rho) = \sum_{D=1}^{D_{\max}} \pi_D \sum_{t=0}^{D-1} \frac{e^{-\rho} \rho^t}{t!}. \quad (42)$$

Now the bound  $B$  in Theorem 4 is a random variable with the probability mass function

$$P(B = D) = \pi_D,$$

for  $D = 1, \dots, D_{\max}$ . Following the same argument in the proof of Theorem 4, one can show that the upper bound for the stable offered load is

$$G \leq \sum_{D=1}^{D_{\max}} \pi_D \left( D - \sum_{\tau=0}^{D-1} (D - \tau) e^{-G \Lambda'(1)} \frac{G \Lambda'(1)^\tau}{\tau!} \right). \quad (43)$$

*Example 2 (Near-Far SIC Decoding):* For the setting with the capacity of near-far SIC decoding, every time slot can be modeled by the  $\phi$ -ALOHA receiver with

$$\phi(n_1, n_2) = \begin{cases} (n_1, n_2) & \text{if } (n_1, n_2) \leq (1, 1) \\ (0, 0) & \text{otherwise.} \end{cases} \quad (44)$$

The near-far SIC decoding is commonly used for modeling power domain NOMA [44], [45]. There are three affine capacity envelopes for the  $\phi$ -ALOHA receiver: (i) the vector  $(1, 0)$  with the bound 1, and (ii) the vector  $(0, 1)$  with the bound 1, and (iii) the vector  $(1, 1)$  with the bound 2. Suppose that there is only one class of receivers ( $J = 1$ ). Then the first affine capacity envelope leads to the bound

$$G_1 \leq 1 - e^{-G_1 \Lambda'_1(1)}, \quad (45)$$

and the second affine capacity envelope leads to the bound

$$G_2 \leq 1 - e^{-G_2 \Lambda'_2(1)}, \quad (46)$$

and the third affine capacity envelope leads to the bound

$$\begin{aligned} G_1 + G_2 &\leq 1 \cdot e^{-(G_1 \Lambda'_1(1) + G_2 \Lambda'_2(1))} \frac{(G_1 \Lambda'_1(1) + G_2 \Lambda'_2(1))}{1!} \\ &\quad + 2 \left( 1 - e^{-(G_1 \Lambda'_1(1) + G_2 \Lambda'_2(1))} \left( 1 + \frac{(G_1 \Lambda'_1(1) + G_2 \Lambda'_2(1))}{1!} \right) \right). \end{aligned} \quad (47)$$

*Example 3 (IRSA System With Two Classes of Users and Two Classes of Receivers):* We consider a system of coded Poisson receivers with two classes of users ( $K = 2$ ), two classes of receivers ( $J = 2$ ), and the success probability function  $P_{\text{suc}}(\rho) = e^{-\rho}$ . This system is referred to as the *IRSA system with two classes of users and two classes of receivers* in [25]. As in [25], we set  $F_1 = F_2 = 0.5$ . We consider the following two packet routing policies (that correspond to two network slicing policies for resource allocation in uplink grant-free transmissions):

- 1) Complete sharing: every packet has an equal probability to be routed to the two classes of receivers, i.e.,  $r_{11} = r_{22} = r_{12} = r_{21} = 0.5$ .
- 2) Receiver reservation: class 1 packets are routed to the two classes of receivers with an equal probability, i.e.,  $r_{11} = r_{12} = 0.5$ , and class 2 packets are routed to the class 2 receivers, i.e.,  $r_{21} = 0, r_{22} = 1$ .

Such a system is a CPR having three capacity envelopes: (i) the vector  $(1, 0)$  with the bound 1, and (ii) the vector  $(0, 1)$  with the bound 1, and (iii) the vector  $(1, 1)$  with the bound 1.

For the complete sharing policy and the affine capacity envelope  $(1, 1)$  with the bound 1, we have from (26) that

$$\mu_1 = \mu_2 = G_1 \Lambda'_1(1) + G_2 \Lambda'_2(1). \quad (48)$$

Hence, by (25), we have the outer bound

$$G_1 + G_2 \leq 1 - e^{-G_1 \Lambda'_1(1) - G_2 \Lambda'_2(1)}. \quad (49)$$

Similarly, for the receiver reservation policy and the affine capacity envelope  $(1, 1)$  with the bound 1, we have from (26) that

$$\mu_1 = G_1 \Lambda'_1(1),$$

$$\mu_2 = G_1 \Lambda'_1(1) + 2G_2 \Lambda'_2(1). \quad (50)$$

Hence, by (25), we have the outer bound

$$G_1 + G_2 \leq 1 - \frac{1}{2} e^{-G_1 \Lambda'_1(1)} - \frac{1}{2} e^{-G_1 \Lambda'_1(1) - 2G_2 \Lambda'_2(1)}. \quad (51)$$

On the other hand, for the receiver reservation policy and the affine capacity envelope  $(0, 1)$  with the bound 1, we have from (26) and (25) that  $\mu_1 = 0, \mu_2 = 2G_2 \Lambda'_2(1)$  and

$$G_2 \leq \frac{1}{2} - \frac{1}{2} e^{-2G_2 \Lambda'_2(1)}. \quad (52)$$

We present the visualization of the bounds (49), (51), and (52) in this example in Figure 2 and Figure 3 in Section VI-B.

#### IV. CONVOLUTIONAL CODED POISSON RECEIVERS

Inspired by the great performance of convolutional LDPC codes (see, e.g., [29], [30], [31]) and the spatially coupled IRSA [27], [32], in this section, we propose using the construction method for convolutional LDPC codes and spatially coupled IRSA to construct convolutional CPRs.

##### A. Circular Convolutional Coded Poisson Receivers

Recall that the ensemble of bipartite graphs in the  $(T, G, \Lambda(x), R, F)$ -CPR is constructed with the following parameters:

- 1) The total number of Poisson receivers  $T$ .
- 2) The offered load vector  $G = (G_1, G_2, \dots, G_K)$ .
- 3) The degree distribution vector  $\Lambda(x) = (\Lambda_1(x), \Lambda_2(x), \dots, \Lambda_K(x))$ .
- 4) The  $K \times J$  routing matrix  $R = (r_{k,j})$ .
- 5) The partition vector  $F = (F_1, F_2, \dots, F_J)$ .

Following the construction of convolutional LDPC codes, we first take  $L$  independent copies of  $(T, G^{(\ell)}, \Lambda(x), R, F)$ -CPR with different offered load vectors,  $G^{(\ell)} = (G_1^{(\ell)}, G_2^{(\ell)}, \dots, G_K^{(\ell)})$ ,  $\ell = 1, 2, \dots, L$ , and concatenate the  $L$  bipartite graphs to form a single bipartite graph. Then we rewire the edges in the concatenated bipartite graph. The receiver end of each edge in the  $\ell^{th}$  copy is rewired to the corresponding receiver node in the  $\hat{\ell}^{th}$  copy, where  $\hat{\ell}$  is chosen uniformly in  $[\ell, \ell \oplus (w-1)]$ , where  $1 \leq w \leq L$  is known as the “smoothing” window size in [29], and the  $\oplus$  operator is the usual addition in a circular manner, i.e., for  $1 \leq \ell \leq L$ ,

$$\ell \oplus (w-1) = \begin{cases} \ell + (w-1) & \text{if } \ell + (w-1) \leq L \\ \ell + (w-1) - L & \text{if } \ell + (w-1) > L. \end{cases} \quad (53)$$

Also, we define the  $\ominus$  operator as the usual subtraction in a circular manner, i.e.,

$$\ell \ominus (w-1) = \begin{cases} \ell - (w-1) & \text{if } \ell - (w-1) \geq 1 \\ \ell - (w-1) + L & \text{if } \ell - (w-1) < 1. \end{cases} \quad (54)$$

We call such a system the circular convolutional CPR with  $L$  stages. By viewing the class  $k$  user nodes (resp. class  $j$  receiver nodes) at the  $\ell$  stage as the class  $(k, \ell)$  users (resp. class  $(j, \ell)$  receivers), the circular convolutional CPR with  $L$  stages is a CPR with  $KL$  classes of users and  $JL$  classes of receivers. Moreover, the routing probability from a  $(k, \ell)$  user node to

a  $(j, \hat{\ell})$  receiver node is  $r_{k,j}/w$  if  $\hat{\ell} \in [\ell, \ell \oplus (w-1)]$  and 0 otherwise. Thus, the density evolution analysis in Theorem 1 can still be applied and we have the following corollary for the circular convolutional CPR with  $L$  stages.

*Corollary 1:* Consider the circular convolutional CPR with  $L$  stages described in this section. Let  $q_{k,\ell}^{(i)}$  be the probability that the *user end* of a randomly selected class  $k$  edge in the  $\ell^{\text{th}}$  stage has not been successfully received after the  $i^{\text{th}}$  SIC iteration. Also, let

$$\rho_{k,j,\ell}^{(i)} = \sum_{\hat{\ell}=\ell \oplus (w-1)}^{\ell} q_{k,\hat{\ell}}^{(i)} G_k^{(\hat{\ell})} \Lambda_k'(1) \frac{1}{w} r_{k,j}/F_j, \quad (55)$$

be the offered load of class  $k$  users to a class  $j$  receiver in the  $\ell^{\text{th}}$  stage after the  $i^{\text{th}}$  SIC iteration, and

$$\tilde{\rho}_{j,\ell}^{(i)} = (\rho_{1,j,\ell}^{(i)}, \rho_{2,j,\ell}^{(i)}, \dots, \rho_{K,j,\ell}^{(i)}), \quad (56)$$

be the offered load vector to a class  $j$  receiver in the  $\ell^{\text{th}}$  stage after the  $i^{\text{th}}$  SIC iteration.

As  $T \rightarrow \infty$ , the success probability for a class  $k$  user in the  $\ell^{\text{th}}$  stage after the  $i^{\text{th}}$  SIC iteration converges to

$$\begin{aligned} \tilde{P}_{\text{suc},k,\ell}^{(i)}(G^{(1)}, G^{(2)}, \dots, G^{(L)}) \\ = 1 - \Lambda_k \left( 1 - \sum_{\hat{\ell}=\ell}^{\ell \oplus (w-1)} \sum_{j=1}^J \frac{1}{w} r_{k,j} P_{\text{suc},k,j}(\tilde{\rho}_{j,\hat{\ell}}^{(i)}) \right), \end{aligned} \quad (57)$$

$k = 1, 2, \dots, K$ ,  $\ell = 1, 2, \dots, L$ , and  $q_{\ell}^{(i)} = (q_{1,\ell}^{(i)}, q_{2,\ell}^{(i)}, \dots, q_{K,\ell}^{(i)})$  can be computed recursively from the following equation:

$$q_{k,\ell}^{(i+1)} = \lambda_k \left( 1 - \sum_{\hat{\ell}=\ell}^{\ell \oplus (w-1)} \sum_{j=1}^J \frac{1}{w} r_{k,j} P_{\text{suc},k,j}(\tilde{\rho}_{j,\hat{\ell}}^{(i)}) \right), \quad (58)$$

with  $q_{\ell}^{(0)} = (1, 1, \dots, 1)$ . Also, under (A4), we can let  $p_{k,\ell}^{(i+1)} = \lambda_k^{-1}(q_{k,\ell}^{(i+1)})$  and  $p_{\ell}^{(i)} = (p_{1,\ell}^{(i)}, p_{2,\ell}^{(i)}, \dots, p_{K,\ell}^{(i)})$ . Then we can rewrite (58) as follows:

$$p_{k,\ell}^{(i+1)} = 1 - \sum_{\hat{\ell}=\ell}^{\ell \oplus (w-1)} \sum_{j=1}^J \frac{1}{w} r_{k,j} P_{\text{suc},k,j}(\tilde{\rho}_{j,\hat{\ell}}^{(i)}), \quad (59)$$

with  $p_{\ell}^{(0)} = (1, 1, \dots, 1)$ .

Similarly, the stability results in Section II-C can also be applied.

### B. Stability and Threshold Saturation

Now we construct the convolutional CPR from the circular convolutional CPR by setting the offered load vectors in the last  $w-1$  stages to be the zero vector  $\mathbf{0}$ . This is known as the *shortening* technique in convolutional LDPC codes.

*Definition 6:* The convolutional  $(T, G, \Lambda(x), R, F, w)$ -CPR with  $L$  stages is the circular convolutional CPR with  $L$  stages when  $G^{(\ell)} = G$  for  $\ell = 1, 2, \dots, L-w+1$ , and  $G^{(\ell)} = \mathbf{0}$  for  $\ell = L-w+2, \dots, L$ .

As the circular convolutional CPR with  $L$  stages is a CPR with  $KL$  classes of users and  $JL$  classes of receivers, the convolutional  $(T, G, \Lambda(x), R, F, w)$ -CPR with  $L$  stages is a

CPR with  $K$  classes of users and  $JL$  classes of receivers (by grouping the  $(k, \ell)$  user nodes at the  $L$  stages into a single class of user nodes, i.e., class  $k$  user nodes). The notion of stability in Section II-C can be defined the same way for the convolutional  $(T, G, \Lambda(x), R, F, w)$ -CPR with  $L$  stages, i.e., it is stable if the probability that a packet is successfully received approaches 1 when the number of iterations goes to infinity.

Note that the normalized offered load for the convolutional  $(T, G, \Lambda(x), R, F, w)$ -CPR with  $L$  stages is

$$\frac{\sum_{\ell=1}^L G^{(\ell)} T}{LT} = \frac{L-w+1}{L} G. \quad (60)$$

This is smaller than the normalized offered load  $G$  of the  $(T, G, \Lambda(x), R, F)$ -CPR. However, as  $L \rightarrow \infty$ , the loss in the normalized offered load for the convolutional  $(T, G, \Lambda(x), R, F, w)$ -CPR with  $L$  stages approaches 0.

One of the most interesting phenomena for convolutional LDPC codes is *threshold saturation* (see, e.g., [29], [31], [32], [33]). In the following theorem, we show an analogous result for the convolutional coded Poisson receivers. As mentioned earlier, the convolutional  $(T, G, \Lambda(x), R, F, w)$ -CPR with  $L$  stages can be viewed as a CPR with  $KL$  classes of users and  $JL$  classes of receivers. As in Definition 3 and Definition 4, its stability region, denoted by  $S_L$ , is defined as the maximal stable set in  $\mathbb{R}_{\geq 0}^K$  such that (i) any  $G \in S_L$  is stable, and (ii) any  $G \notin S_L$  is not stable.

*Theorem 5 (Threshold Saturation):* Let  $S$  be the stability region of the  $(T, G, \Lambda(x), R, F)$ -CPR and  $S_L$  be the stability region of the convolutional  $(T, G, \Lambda(x), R, F, w)$ -CPR with  $L$  stages. Then for any positive integer  $L$ ,

$$S \subset S_L \subset S_{L-1}. \quad (61)$$

In view of Theorem 5, the stability region  $S_L$  of the convolutional  $(T, G, \Lambda(x), R, F, w)$ -CPR with  $L$  stages “saturates” when  $L \rightarrow \infty$ .

*Proof:* Recall that the convolutional  $(T, G, \Lambda(x), R, F, w)$ -CPR with  $L$  stages is the circular convolutional Poisson receiver with  $L$  stages when the last  $w-1$  stages are “shortened,” i.e.,  $G^{(\ell)} = G$  for  $\ell = 1, 2, \dots, L-w+1$ , and  $G^{(\ell)} = \mathbf{0}$  for  $\ell = L-w+2, \dots, L$ . If the last  $w$  stages are “shortened,” i.e.,  $G^{(\ell)} = G$  for  $\ell = 1, 2, \dots, L-w$ , and  $G^{(\ell)} = \mathbf{0}$  for  $\ell = L-w+1, \dots, L$ , then no packets are sent to the last stage and it can be removed. Thus, it reduces to the convolutional  $(T, G, \Lambda(x), R, F, w)$ -CPR with  $L-1$  stages. As the success probability functions are decreasing in the offered load in (A1), the success probability in the convolutional  $(T, G, \Lambda(x), R, F, w)$ -CPR with  $L$  stages is not larger than that of the convolutional  $(T, G, \Lambda(x), R, F, w)$ -CPR with  $L-1$  stages. This shows that  $S_L \subset S_{L-1}$ .

Now consider the circular convolutional Poisson receiver with  $L$  stages when no stages are “shortened,” i.e.,  $G^{(\ell)} = G$  for  $\ell = 1, 2, \dots, L$ . From the monotonicity, it is clear that the success probability of this system is not larger than that of the convolutional  $(T, G, \Lambda(x), R, F, w)$ -CPR with  $L$  stages. We will argue that the success probability of this system is exactly the same as that of the  $(T, G, \Lambda(x), R, F)$ -CPR. As such, we have  $S \subset S_L$ .



When  $G^{(\ell)} = G$  for  $\ell = 1, 2, \dots, L$ ,  $\rho_{k,j,\ell}^{(i)}$  in (55) can be simplified as follows:

$$\rho_{k,j,\ell}^{(i)} = G_k \Lambda'_k(1) r_{k,j} / F_j \sum_{\hat{\ell}=\ell \ominus (w-1)}^{\ell} q_{k,\hat{\ell}}^{(i)}. \quad (62)$$

We now use induction to show that  $q_{k,\ell}^{(i)} = q_k^{(i)}$  for all  $i$ . Since  $q_{k,\ell}^{(0)} = 1$  and  $q_k^{(0)} = 1$ , we have from (62) that

$$\rho_{k,j,\ell}^{(1)} = G_k \Lambda'_k(1) r_{k,j} / F_j. \quad (63)$$

In view of (13) and (56), we have

$$\tilde{\rho}_{j,\ell}^{(1)} = q_{\ell}^{(1)} \circ \tilde{\rho}_j.$$

It is easy to see from (58) and (15) that  $q_{k,\ell}^{(1)} = q_k^{(1)}$  for all  $\ell = 1, 2, \dots, L$ . By inducting on the number of iterations  $i$ , we then have  $q_{k,\ell}^{(i)} = q_k^{(i)}$  for all  $i$ . Because of (57) and (14), we conclude that the success probability of the circular convolutional Poisson receiver without shortening is the same as that of the  $(T, G, \Lambda(x), R, F)$ -CPR. ■

In the case where there is only one class of users and one class of receivers (the single-class system), the stability region of the convolutional  $(T, G, \Lambda(x), R, F, w)$ -CPR with  $L$  stages can be reduced to a set in  $\mathbb{R}_{\geq 0}$ . The supremum of this set represents the percolation threshold of the CCPR. A direct finding from Theorem 5 is the monotonic decrease of the percolation threshold as the number of stages  $L$  increases.

*Example 4 (Convolutional IRSA):* For slotted ALOHA, the success probability function  $P_{\text{suc}}(\rho) = e^{-\rho}$ . Suppose that the degree distribution  $\Lambda(x) = x^d$ , i.e., each packet is transmitted exactly  $d$  times. For such a degree distribution, we have  $\lambda(x) = x^{d-1}$ . Assume  $w = 2$ ,  $K = 1$ , and  $J = 1$ . It follows from (58) in Corollary 1 that for  $\ell = 2, \dots, L-2$ ,

$$q_1^{(i+1)} = \left( 1 - \frac{1}{2} e^{-\frac{1}{2} G d q_1^{(i)}} - \frac{1}{2} e^{-\frac{1}{2} G d (q_1^{(i)} + q_2^{(i)})} \right)^{d-1}, \quad (64)$$

$$q_{\ell}^{(i+1)} = \left( 1 - \frac{1}{2} e^{-\frac{1}{2} G d (q_{\ell-1}^{(i)} + q_{\ell}^{(i)})} - \frac{1}{2} e^{-\frac{1}{2} G d (q_{\ell}^{(i)} + q_{\ell+1}^{(i)})} \right)^{d-1}, \quad (65)$$

and

$$q_{L-1}^{(i+1)} = \left( 1 - \frac{1}{2} e^{-\frac{1}{2} G d (q_{L-2}^{(i)} + q_{L-1}^{(i)})} - \frac{1}{2} e^{-\frac{1}{2} G d q_{L-1}^{(i)}} \right)^{d-1}, \quad (66)$$

with the initial condition  $q_k^{(0)} = 1$ ,  $k = 1, 2, \dots, L-1$ . It can be shown that from Example 5 of [25] that the percolation thresholds are  $G = 0.9179, 0.9767, 0.9924, 0.9973$  for  $d = 3, 4, 5, 6$ . We evaluate the thresholds for the system described in this example with  $w = 2, 3, 4$  in Section VI-A.

## V. PERCOLATION THRESHOLDS OF SYSTEMS WITH A SINGLE CLASS OF USERS

In this section, we focus on CPR systems with only one class of users and one class of receivers. We derive the potential function of such a system of CPR and the associated percolation thresholds of both CPRs and CCPRs.

### A. The Single-System Threshold

Since there is only one class of users and one class of receivers, (15) and (16) reduce to the following form:

$$p^{(i)} = 1 - P_{\text{suc}}(q^{(i)} G \Lambda'(1)), \quad (67)$$

$$q^{(i+1)} = \lambda(p^{(i)}). \quad (68)$$

We combine the two equations in (67) and (68) to form the following recursive equation:

$$p^{(i+1)} = 1 - P_{\text{suc}}(\lambda(p^{(i)}) G \Lambda'(1)), \quad (69)$$

with  $p^{(0)} = 1$ . For every packet to be decoded successfully, we need to ensure that

$$\lim_{i \rightarrow \infty} p^{(i)} = 0. \quad (70)$$

We are interested in finding out the maximum offered load  $G$  such that (70) is satisfied. Let

$$f(p; G) = 1 - P_{\text{suc}}(p G \Lambda'(1)), \quad (71)$$

$$h(p) = \lambda(p). \quad (72)$$

Then (69) can be written as follows:

$$p^{(i+1)} = f(h(p^{(i)}); G). \quad (73)$$

In [30], the recursion of the form in (73) is said to be a *scalar admissible system* characterized by a pair of functions  $(f, h)$  that satisfy the four properties in Definition 7 below.

*Definition 7 (cf. Def. 1 in [30]):* The scalar admissible system  $(f, h)$  parameterized by  $G \geq 0$  is defined by the recursion  $p^{(i+1)} = f(h(p^{(i)}); G)$ , where  $f$  and  $h$  satisfy the following four properties:

- (P1)  $f : [0, 1] \times [0, \infty) \rightarrow [0, 1]$  is strictly increasing in both  $p$  and  $G$ .
- (P2)  $h : [0, 1] \rightarrow [0, 1]$  satisfies  $h'(p) > 0$  for  $p \in (0, 1]$ .
- (P3)  $f(0; G) = f(p; 0) = h(0) = 0$ .
- (P4)  $f$  has continuous second derivatives on  $[0, 1] \times [0, \infty)$  w.r.t. all arguments, so is  $h$  on  $[0, 1]$ .

In the following lemma, we show that the  $(T, G, \Lambda(x), R, F)$ -CPR with one class of users and one class of receivers is indeed a scalar admissible system under the assumptions in (A1)-(A4).

*Lemma 2:* Under the assumptions in (A1) and (A4), a system of CPR described by the density evolution equation in (69) is the scalar admissible system  $(f, h)$  with  $f$  and  $h$  specified in (71) and (72), respectively.

*Proof:* It suffices to show that  $f$  in (71) and  $h$  in (72) satisfy (P1)-(P4) of Definition 7.

(P1) Since  $P_{\text{suc}}$  is (strictly) decreasing in (A1), we have for  $p \in [0, 1]$  and  $G \in [0, \infty)$ ,

$$\frac{\partial f(p; G)}{\partial p} = - \frac{\partial P_{\text{suc}}(p G \Lambda'(1))}{\partial p} \cdot G \Lambda'(1) > 0,$$

$$\frac{\partial f(p; G)}{\partial G} = - \frac{\partial P_{\text{suc}}(p G \Lambda'(1))}{\partial G} \cdot p \Lambda'(1) > 0.$$

(P2) Since every packet is transmitted at least twice in (A4), we have  $\Lambda''(p) > 0$ ,  $\forall p \in [0, 1]$ . Also, since  $\Lambda'(1)$  is positive,  $h'(p) = \lambda'(p) = \Lambda''(p)/\Lambda'(1) > 0$ .

(P3) Since  $P_{\text{suc}}(0) = 1$ , and  $\lambda$  is a polynomial without constant terms,

$$\begin{aligned} f(0, G) &= 1 - P_{\text{suc}}(0) = 0, \\ f(p, 0) &= 1 - P_{\text{suc}}(0) = 0, \\ h(0) &= \lambda(0) = 0. \end{aligned}$$

(P4) is obvious from (A1). ■

One of the most powerful tools for analyzing the stability of the scalar admissible system  $(f, h)$  is the *potential function* in [30].

**Definition 8:** The function  $U(p; G)$  is called the *potential function* of the scalar admissible system  $(f, h)$  in [30] if it satisfies

$$\begin{aligned} p^{(i+1)} - p^{(i)} &= f(h(p^{(i)}); G) - p^{(i)} \\ &= -\frac{1}{h'(p^{(i)})} \frac{\partial U(p; G)}{\partial p} \Big|_{p=p^{(i)}}. \end{aligned} \quad (74)$$

To ease the representation, for a function  $U(p; G)$  with two variables, we use  $U'(p; G)$  to denote the partial differentiation with respect to the *first* variable, given by

$$U'(p; G) = \frac{\partial U(p; G)}{\partial p}.$$

Rewrite (74) as follows:

$$U'(p; G) \Big|_{p=p^{(i)}} = -h'(p^{(i)})(f(h(p^{(i)}); G) - p^{(i)}). \quad (75)$$

Integrating both sides of (75) yields

$$\begin{aligned} U(p; G) &= \int_0^p h'(z)(z - f(h(z); G)) dz \\ &= ph(p) - H(p) - F(h(p); G), \end{aligned} \quad (76)$$

where

$$F(p; G) = \int_0^p f(z; G) dz, \quad (77)$$

and

$$H(p) = \int_0^p h(z) dz. \quad (78)$$

To see how the potential function can be used for analyzing the stability of a scalar admissible system, let us consider the *single-system threshold* [30] defined below:

$$G_s^* = \sup\{G \in [0, 1] | U'(p; G) > 0, \forall p \in (0, 1]\}. \quad (79)$$

Then for all  $G \leq G_s^*$ , the system is stable. To see this, note from (P1) of Definition 7 that  $U'(p; G) > 0$  for all  $p \in (0, 1]$  and  $G \leq G_s^*$ . Also, note from (P3) of Definition 7, that is,  $f(0; G) = f(p; 0) = h(0) = 0$ , that  $p = 0$  is a solution of  $p = f(h(p); G)$ . Since  $h'(p) > 0$  for  $p \in (0, 1]$  ((P2) in Definition 7), one can see from (74) that  $p^{(i+1)} < p^{(i)}$  for all  $i$ . Thus,  $p^{(i)}$  (with  $p^{(0)} = 1$ ) converges to 0 as  $i \rightarrow \infty$  for all  $G \leq G_s^*$ .

Since we have shown in Lemma 2 that the  $(T, G, \Lambda(x), R, F)$ -CPR with one class of users and one class of receivers is a scalar admissible system with  $f$  and  $h$  specified in (71) and (72), we can use (76) to derive the potential function  $U(p; G)$  for  $(T, G, \Lambda(x), R, F)$ -CPRs, and then use that to show the following stability result.

**Theorem 6:** Under the assumptions in (A1)-(A4), the potential function of the  $(T, G, \Lambda(x), R, F)$ -CPR with one class of users and one class of receivers is given by

$$\begin{aligned} U(p; G) &= \lambda(p)(p-1) - \frac{\Lambda(p)}{\Lambda'(1)} \\ &\quad + \frac{1}{G\Lambda'(1)} \int_0^{G\Lambda'(1)\lambda(p)} P_{\text{suc}}(\rho) d\rho. \end{aligned} \quad (80)$$

Moreover, if the inverse function of  $P_{\text{suc}}(\cdot)$  exists and is continuous and decreasing on  $[0, 1]$ , then for all  $G \leq G_s^*$ , the system is stable, where  $G_s^*$  in (79) has the following representation:

$$G_s^* = \inf_{p \in [0, 1]} \frac{P_{\text{suc}}^{-1}(1-p)}{\Lambda'(p)}. \quad (81)$$

**Proof:** In view of (71) and (77), integrating the function  $f$  w.r.t.  $z$  from 0 to  $p$  yields

$$\begin{aligned} F(p; G) &= \int_0^p 1 - P_{\text{suc}}(zG\Lambda'(1)) dz \\ &= p - \frac{1}{G\Lambda'(1)} \int_0^{G\Lambda'(1)p} P_{\text{suc}}(\rho) d\rho. \end{aligned} \quad (82)$$

Also, note that  $h(p) = \lambda(p) = \Lambda'(p)/\Lambda'(1)$ ,  $H(p) = \Lambda(p)/\Lambda'(1)$ . Using (76), we obtain (80).

From (74), the partial derivative of the potential function w.r.t. its first variable, known as the *balance function* has the following form:

$$U'(p; G) = \lambda'(p)(p-1 + P_{\text{suc}}(G\Lambda'(p))), \quad (83)$$

where  $\Lambda'(p) = \Lambda'(1)\lambda(p)$ . Since we assume that  $P_{\text{suc}}^{-1}$  exists, from (79) and (83), we have that

$$\begin{aligned} G_s^* &= \sup\{G \in [0, \infty) | U'(p; G) > 0, \forall p \in (0, 1]\} \\ &= \sup\{G \in [0, \infty) | P_{\text{suc}}(G\Lambda'(p)) > 1-p, \\ &\quad \forall p \in (0, 1]\} \\ &= \inf_{p \in [0, 1]} \frac{P_{\text{suc}}^{-1}(1-p)}{\Lambda'(p)}. \end{aligned} \quad (84)$$

■

The following example shows the potential functions of the  $D$ -fold ALOHA system.

**Example 5 ( $D$ -Fold ALOHA):** For the  $D$ -fold ALOHA system described by (3) and (5) with  $\Lambda(x) = x^d$ , we use (80) and integration by parts (as in Theorem 3 of [46]) to evaluate the potential function as follows:

$$\begin{aligned} U(p; G) &= \frac{1}{d} \left( (d-1)p^d - dp^{d-1} + \frac{1}{G} \left( D \right. \right. \\ &\quad \left. \left. - e^{-Gdp^{d-1}} \sum_{\tau=0}^{D-1} (D-\tau) \frac{(Gdp^{d-1})^\tau}{\tau!} \right) \right). \end{aligned} \quad (85)$$

## B. The Saturation Theorem

In this section, we study the stability of the convolutional  $(T, G, \Lambda(x), R, F, w)$ -CPR with  $L$  stages. The density evolution equation for the convolutional  $(T, G, \Lambda(x), R, F, w)$ -CPR

is given by (59) in Corollary 1. For  $K = J = 1$ , by (11), (55), and the relation  $q_\ell^{(i)} = \lambda(p_\ell^{(i)})$ , (59) can be simplified as

$$p_\ell^{(i+1)} = 1 - \sum_{\tilde{\ell}=\ell}^{\ell \oplus (w-1)} \frac{1}{w} P_{\text{suc}} \left( \frac{1}{w} \sum_{\tilde{\ell}=\ell \ominus (w-1)}^{\tilde{\ell}} \lambda(p_{\tilde{\ell}}^{(i)}) G^{(\tilde{\ell})} \Lambda'(1) \right), \quad (86)$$

for  $\ell = 1, \dots, L$ . Moreover, since  $G^{(\tilde{\ell})} = G$  for  $\tilde{\ell} = 1, 2, \dots, L - w + 1$  and  $G^{(\tilde{\ell})} = 0$  for  $\tilde{\ell} = L - w + 2, \dots, L$ , (86) can be further simplified as follows:

$$p_\ell^{(i+1)} = 1 - \sum_{\tilde{\ell}=\ell}^{\ell \oplus (w-1)} \frac{1}{w} P_{\text{suc}} \left( \frac{G \Lambda'(1)}{w} \sum_{\tilde{\ell}=\max[1, \ell - (w-1)]}^{\min[L - w + 1, \tilde{\ell}]} \lambda(p_{\tilde{\ell}}^{(i)}) \right), \quad (87)$$

for  $\ell = 1, 2, \dots, L - w + 1$ .

Denote by  $G_{\text{conv}}^*(L, w)$  the percolation threshold of the convolutional  $(T, G, \Lambda(x), R, F, w)$ -CPR system with  $L$  stages. The only way to know the exact value of  $G_{\text{conv}}^*(L, w)$  is to evaluate the density evolution equations in (86). In this section, we prove a lower bound of  $G_{\text{conv}}^*(L, w)$  by using the potential function of the corresponding  $(T, G, \Lambda(x), R, F)$ -CPR (under certain conditions of the window size  $w$ ). Such a stability result for the convolutional  $(T, G, \Lambda(x), R, F, w)$ -CPR system with  $L$  stages is called the *saturation theorem* in this paper.

In addition to the single-system threshold  $G_s^*$  in (79), we define the *potential threshold* of the  $(T, G, \Lambda(x), R, F)$ -CPR. We will show that the potential threshold is a lower bound of  $G_{\text{conv}}^*(L, w)$  in the saturation theorem.

**Definition 9** (cf. Def. 6 in [30]): Consider the scalar admissible system described in Definition 7 with the potential function  $U(p; G)$  in (76). The potential threshold of the scalar admissible system, denoted by  $G_{\text{conv}}^*$ , is defined below:

$$G_{\text{conv}}^* = \sup\{G \in [0, 1] \mid \min_{p \in [0, 1]} U(p; G) \geq 0\}. \quad (88)$$

To give the specific conditions for the window size of the saturation theorem, we define the following terms.

**Definition 10** (cf. Def. 5, and 6 in [30]): Consider the scalar admissible system  $(f, h)$  in Definition 7 with the potential function  $U(p; G)$  in (76).

- (i) For  $G > G_s^*$ , the minimum unstable fixed point is the number

$$u(G) = \sup\{\tilde{p} \in [0, 1] \mid U'(\tilde{p}; G) \geq 0, \tilde{p} \in [0, \tilde{p}]\}. \quad (89)$$

- (ii) The energy gap of the scalar admissible system for  $G \in (G_s^*, 1]$  is the number

$$\Delta E(G) = \min_{p \in [u(G), 1]} U(p; G). \quad (90)$$

- (iii) A constant  $K_{f,h} = \|h'\|_\infty + \|h'\|_\infty^2 \|f'\|_\infty + \|h''\|_\infty$ , where

$$\|h\|_\infty = \sup_{x \in [0, 1]} |h(x)|$$

for functions  $h : [0, 1] \rightarrow \mathbb{R}$ .

Now we state the saturation theorem.

**Theorem 7 (The Saturation Theorem for the Convolutional  $(T, G, \Lambda(x), R, F, w)$ -CPR):** Consider the single-class ( $K = J = 1$ ) convolutional  $(T, G, \Lambda(x), R, F, w)$ -CPR system with

$L$  stages governed by the density evolution recursion in (87). If  $G < G_{\text{conv}}^*$  and

$$w > \frac{K_{f,h}}{\Delta E(G)}, \quad (91)$$

then the only fixed point of (87) is  $\mathbf{p} = \mathbf{0}$ . As such, the convolutional  $(T, G, \Lambda(x), R, F, w)$ -CPR system with  $L$  stages is stable for all  $G < G_{\text{conv}}^*$ .

*Proof:* Let  $\tilde{L} = L - w + 1$ . We reverse all the indices of the recursion relation (87). Then, the recursion (87) becomes

$$p_\ell^{(i+1)} = 1 - \sum_{\tilde{\ell}=\tilde{L}+1-\ell}^{\tilde{L}+w-\ell} \frac{1}{w} P_{\text{suc}} \left( \frac{G \Lambda'(1)}{w} \sum_{\tilde{\ell}=\max[1, \tilde{L}-\ell+1]}^{\min[\tilde{L}, \tilde{L}-\ell+w]} \lambda(p_{\tilde{\ell}}^{(i)}) \right). \quad (92)$$

If  $h$  is a function with one variable and  $\mathbf{p} = (p_1, \dots, p_{\tilde{L}}) \in \mathbb{R}^{\tilde{L}}$  is a vector, we define

$$\mathbf{h}(\mathbf{p}) = (h(p_1), h(p_2), \dots, h(p_{\tilde{L}}))$$

and

$$\mathbf{f}(\mathbf{p}; G) = (f(p_1; G), f(p_2; G), \dots, f(p_{\tilde{L}}; G))$$

for the ease of representation. Let  $\mathbf{p}^{(i)} = (p_1^{(i)}, p_2^{(i)}, \dots, p_{\tilde{L}}^{(i)})$ , then (92) can be written as the following matrix form:

$$\mathbf{p}^{(i+1)} = \mathbf{A}_2 \mathbf{f}(\mathbf{A}_2^T \mathbf{h}(\mathbf{p}^{(i)}); G), \quad (93)$$

where  $\mathbf{A}_2$  is an  $\tilde{L} \times L$  matrix defined as:

$$\mathbf{A}_2 = \frac{1}{w} \begin{bmatrix} 1 & 1 & \dots & 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 1 & \dots & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & 1 & 1 & \dots & 1 & 0 \\ 0 & \dots & 0 & 0 & 1 & 1 & \dots & 1 \end{bmatrix}, \quad (94)$$

where all the rows of  $\mathbf{A}_2$  contain  $w$  ones, and the functions  $f$  and  $h$  are specified in (71) and (72), respectively. We observe that (93) is actually the matrix form of equation (1) in [35] and  $\mathbf{A}_2$  equals matrix  $\mathbf{A}$  in equation (2) in [35]. This theorem thus follows directly from Lemma 29 in [35]. ■

Note that if  $L$  is large enough, one can always choose a window size  $w$  large enough to satisfy (91). An immediate consequence of Theorem 7 is that  $G_{\text{conv}}^*(L, w) \geq G_{\text{conv}}^*$  if  $w > K_{f,h}/\Delta E(G)$ , as Theorem 7 provides a sufficient condition for the stability of the convolutional  $(T, G, \Lambda(x), R, F, w)$ -CPR system with  $L$  stages. However, it is also possible that  $G_{\text{conv}}^*(L, w)$  is larger than  $G_{\text{conv}}^*$  even when the condition in (91) is not satisfied. This is due to the shortening effect that reduces the offered load. In addition, the condition for  $w$  in Theorem 7 often results in large window size. However, our numerical results in Section VI-A demonstrate that  $|G_{\text{conv}}^*(L, w) - G_{\text{conv}}^*| < 0.0001$  can be achieved in some cases. Furthermore, as pointed out by one of the reviewers, one can use the results in [35] to show the following theorem.

**Theorem 8:** Consider the single-class ( $K = J = 1$ ) convolutional  $(T, G, \Lambda(x), R, F, w)$ -CPR system with  $L$  stages governed by the density evolution recursion in (87). Then

$$\lim_{w \rightarrow \infty} \lim_{L \rightarrow \infty} G_{\text{conv}}^*(L, w) = G_{\text{conv}}^*. \quad (95)$$

*Proof:* For any  $G < G_{conv}^*$ , threshold saturation (Theorem 7) implies that there is a finite  $w$  such that the convolutional  $(T, G, \Lambda(x), R, F, w)$ -CPR system with  $L$  stages is stable for all  $L$ . Thus, Theorem 5 implies that the limit in (95) is greater or equal to  $G_{conv}^*$ .

On the other hand, consider the  $\bar{p}^*$  such that

$$U(\bar{p}^*; G_{conv}^*) = \Delta E(G_{conv}^*) = 0.$$

Then, for some  $L, w > 0$ , if  $\Delta E(G^*(L, w)) > 0$ , then

$$\lim_{i \rightarrow \infty} p_\ell^{(i)} \leq \bar{p}^*.$$

By Remark 4 of [35], when applied to this case,

$$\lim_{w \rightarrow \infty} \lim_{L \rightarrow \infty} \lim_{i \rightarrow \infty} p_\ell^{(i)}(L, w) \leq \bar{p}^*.$$

Hence,

$$\Delta E \left( \lim_{w \rightarrow \infty} \lim_{L \rightarrow \infty} G_{conv}^*(L, w) \right) \geq \Delta E(G_{conv}^*) = 0.$$

By Definition 9 and the monotonicity of the potential function in  $G$  ((P1) in Definition 7), the limit in (95) is smaller or equal to  $G_{conv}^*$ . Together, these results establish the stated limit in (95). ■

### C. The Upper Bound by the Area Theorem

In this section, we consider another threshold  $G_{up}^*$  that is defined as the unique positive solution of the equation  $U(1; G) = 0$ . For the potential function of a system of CPR in (80),  $G_{up}^*$  is the unique positive solution to the following equation:

$$G = \int_0^{G\Lambda'(1)} P_{\text{suc}}(\rho) d\rho. \quad (96)$$

Now we show by the area theorem [26] that  $G_{up}^*$  is an upper bound for the stable offered load of a system of CPR. For  $p^{(i)}$  and  $q^{(i)}$  in the two recursive mappings in (67) and (68) to converge to 0 as  $i \rightarrow \infty$ , the area theorem in [26] provides a necessary condition that requires the sum of the two areas under the two mappings to be not greater than 1, i.e.,

$$\int_0^1 (1 - P_{\text{suc}}(qG\Lambda'(1))) dq + \int_0^1 \lambda(p) dp \leq 1. \quad (97)$$

Since  $\lambda(p) = \Lambda'(p)/\Lambda'(1)$ ,  $\int_0^1 \lambda(p) dp = 1/\Lambda'(1)$ . Let  $\rho = qG\Lambda'(1)$ . We can write the inequality in (97) as follows:

$$G \leq \int_0^{G\Lambda'(1)} P_{\text{suc}}(\rho) d\rho. \quad (98)$$

For the  $D$ -fold ALOHA system described by (3) and (5),  $G_{up}^*$  is the solution to the following equation:

$$G = D - \sum_{\tau=0}^{D-1} (D - \tau) e^{-Gd} \frac{(Gd)^\tau}{\tau!}. \quad (99)$$

This is the same as the upper bound in Theorem 1 of [27] for the spatially coupled  $D$ -fold ALOHA system. It is also the same as the upper bound in Section III when the underlying channel is modeled by a  $\phi$ -ALOHA receiver [28]. In view of the extended upper bound for a mixture of  $D$ -fold ALOHA in Example 1, we have the following corollary.

*Corollary 2:* Suppose that the  $T$  Poisson receivers of the  $(T, G, \Lambda(x), R, F)$ -CPR with one class of users and one class of receivers in (80) are induced from a mixture of  $D$ -fold ALOHA in Example 1. Then the stable offered load of such a system is bounded above by  $G_{up}^*$ , where  $G_{up}^*$  is the solution to (96). As such,  $G_{up}^*$  is an upper bound for the percolation threshold of the convolutional  $(T, G, \Lambda(x), R, F, w)$ -CPR with an infinite number of stages, i.e.,

$$\lim_{L \rightarrow \infty} G_{conv}^*(L, w) < G_{up}^*$$

regardless of the choice of the degree distribution  $\Lambda(x)$  and the window size  $w$ .

In the following theorem, we show the single-system threshold  $G_s^*$  in (79) is smaller than the potential threshold  $G_{conv}^*$ , and the potential threshold  $G_{conv}^*$  is smaller than the upper bound  $G_{up}^*$ .

*Theorem 9:* Consider the scalar admissible system  $(f, h)$  in Definition 7 with the potential function  $U(p; G)$ . Then

$$G_s^* < G_{conv}^* < G_{up}^*. \quad (100)$$

*Proof:* (i) First, we demonstrate that  $G_s^* < G_{conv}^*$ . From the integral representation in (76) and the fact that the function  $f$  is strictly increasing in  $G$  from (P1), we deduce that  $U(0; G) = 0$  for all  $G$  and  $U(p; G)$  is strictly decreasing in  $G$ . From the definition given in (79) of  $G_s^*$ , it is clear that  $U'(p; G) > 0$  when  $G \leq G_s^*$ . Therefore, the function  $U(p; G)$  is strictly increasing in  $p$  when  $G \leq G_s^*$ . This implies that  $U(p; G) > 0$  for all  $p \in (0, 1)$  and  $G \leq G_s^*$ . Since  $U(p; G)$  decreases for  $G$ , when  $U(p; G)$  becomes 0 for some value of  $G$  and  $p$  is not 0, such a  $G$  must be greater than  $G_s^*$ . The supremum of all such possible  $G$  values results in  $G_s^* < G_{conv}^*$ .

(ii) Now we show that  $G_{conv}^* < G_{up}^*$ . By (A4), we have  $\Lambda'(1) > 0$  and  $\lambda'(1) > 0$ . Since  $P_{\text{suc}}(G\Lambda'(1)) > 0$  from (A1), we have

$$U'(1; G) = \lambda'(1) P_{\text{suc}}(G\Lambda'(1)) > 0.$$

Thus,  $p = 1$  cannot be a local minimum of  $U(p; G)$  for  $G > 0$ . As such,

$$U(1; G_{conv}^*) > \min_{p \in [0, 1]} U(p; G_{conv}^*) = 0. \quad (101)$$

As argued in (i),  $U(1; G)$  is strictly decreasing in  $G$ . Thus, we have from (101) that  $U(1; G) > 0$  for all  $G \leq G_{conv}^*$ . This shows that  $G_{conv}^* < G_{up}^*$  if  $G_{up}^*$  exists. It remains to show the existence of the unique positive solution  $G_{up}^*$  of the equation  $U(1; G) = 0$ . Since  $U(1; G)$  is continuous and decreasing for  $G > 0$ , and  $f(p; G)$  is strictly increasing in both  $p$  and  $G$ ,  $U(1; G) \rightarrow -\infty$  when  $G \rightarrow \infty$  by (76). Therefore, combining with (101), the unique positive solution to the equation  $U(1; G) = 0$  exists on  $(G_{conv}^*, \infty)$ . ■

Given Theorem 7 and Theorem 9, one can improve the percolation threshold of a  $(T, G, \Lambda(x), R, F)$ -CPR system from  $G_s^*$  to at least  $G_{conv}^*$  by adopting the spatially-coupled method. Though the window size  $w$  needs to be large to satisfy the condition in (91), our numerical results in Section VI-A show that adopting  $w = 2$  significantly improves the percolation threshold.



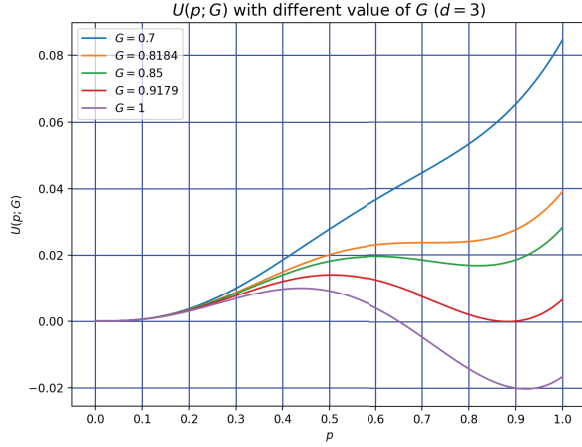


Fig. 1. The plot of the potential function  $U(p; G)$  as a function of  $p$  for various values of  $G$  in Example 6 with  $d = 3$ .

Theorem 9 also reveals the existence of a numerical gap between  $G_{conv}^*$  and  $G_{up}^*$ . However, as shown in Section VI, this gap is small in most cases, with  $|G_{up}^* - G_{conv}^*|$  typically being smaller than  $10^{-2}$ .

*Example 6 (IRSA With a Regular Degree):* In this example, we illustrate how  $G_s^*$ ,  $G_{conv}^*$  and  $G_{up}^*$  are related for IRSA with a regular degree  $d$ , i.e.,  $\Lambda(x) = x^d$  and  $P_{suc}(\rho) = e^{-\rho}$ . Then we have from (85) (with  $D = 1$ ) that

$$U(p; G) = \frac{1}{d} \left( (d-1)p^d - dp^{d-1} + \frac{1 - e^{-Gdp^{d-1}}}{G} \right), \quad (102)$$

and

$$U'(p; G) = (d-1)p^{d-2}(p-1 + e^{-Gdp^{d-1}}). \quad (103)$$

For  $d = 3$ , we use a computer search to find  $G_s^*$ ,  $G_{conv}^*$  and  $G_{up}^*$  as follows:

$$\begin{aligned} G_s^* &= \sup \{ G \in [0, 1] \mid \\ &\quad d(d-1)p^{d-2}(p-1 + e^{-Gdp^{d-1}}) > 0, \forall p \in (0, 1] \} \\ &\approx 0.8184, \\ G_{conv}^* &= \sup \{ G \in [0, 1] \mid \\ &\quad \min_{p \in [0, 1]} (d-1)p^d - dp^{d-1} + \frac{1 - e^{-Gdp^{d-1}}}{G} \geq 0 \} \\ &\approx 0.9179, \\ G_{up}^* &= \{ G : G = 1 - e^{-dG} \} \approx 0.9405. \end{aligned}$$

In Figure 1, we plot the potential function  $U(p; G)$  as a function of  $p$  for various values of  $G$ . This plot illustrates how the potential function changes for the parameter  $p$  under various values of the parameter  $G$ .

## VI. NUMERICAL RESULTS

In this section, we provide numerical results for various systems of convolutional coded Poisson receivers. The convergence criteria are defined as  $|q_{k,\ell}^{(i)} - q_{k,\ell}^{(i-1)}| < 10^{-8}$  (or

TABLE I  
PERCOLATION THRESHOLDS FOR CONVOLUTIONAL IRSA

| Convolutional IRSA |         |         |         |         |         |              |            |
|--------------------|---------|---------|---------|---------|---------|--------------|------------|
| $d$                | $w = 1$ | $w = 2$ | $w = 3$ | $w = 4$ | $G_s^*$ | $G_{conv}^*$ | $G_{up}^*$ |
| 3                  | 0.8184  | 0.9177  | 0.9179  | 0.9179  | 0.8184  | 0.9179       | 0.9405     |
| 4                  | 0.7722  | 0.9708  | 0.9767  | 0.9767  | 0.7722  | 0.9767       | 0.9802     |
| 5                  | 0.7017  | 0.9625  | 0.9914  | 0.9924  | 0.7017  | 0.9924       | 0.9930     |
| 6                  | 0.6370  | 0.9258  | 0.9917  | 0.9970  | 0.6370  | 0.9973       | 0.9975     |

TABLE II  
PERCOLATION THRESHOLDS FOR THE CONVOLUTIONAL 2-FOLD ALOHA

| Convolutional 2-fold ALOHA |         |         |         |         |         |              |            |
|----------------------------|---------|---------|---------|---------|---------|--------------|------------|
| $d$                        | $w = 1$ | $w = 2$ | $w = 3$ | $w = 4$ | $G_s^*$ | $G_{conv}^*$ | $G_{up}^*$ |
| 3                          | 1.5528  | 1.9560  | 1.9760  | 1.9763  | 1.5528  | 1.9764       | 1.9790     |
| 4                          | 1.3336  | 1.8966  | 1.9894  | 1.9961  | 1.3336  | 1.9964       | 1.9966     |
| 5                          | 1.1577  | 1.7722  | 1.9639  | 1.9955  | 1.1577  | 1.9994       | 1.9995     |
| 6                          | 1.0216  | 1.6326  | 1.9071  | 1.9833  | 1.0216  | 1.9999       | 1.9999     |

TABLE III  
PERCOLATION THRESHOLDS FOR THE CONVOLUTIONAL 3-FOLD ALOHA

| Convolutional 3-fold ALOHA |         |         |         |         |         |              |            |
|----------------------------|---------|---------|---------|---------|---------|--------------|------------|
| $d$                        | $w = 1$ | $w = 2$ | $w = 3$ | $w = 4$ | $G_s^*$ | $G_{conv}^*$ | $G_{up}^*$ |
| 3                          | 2.1744  | 2.8990  | 2.9874  | 2.9916  | 2.1744  | 2.9918       | 2.9923     |
| 4                          | 1.8108  | 2.7127  | 2.9577  | 2.9950  | 1.8108  | 2.9993       | 2.9994     |
| 5                          | 1.5456  | 2.4744  | 2.8636  | 2.9739  | 1.5456  | 2.9999       | 3.0000     |
| 6                          | 1.3487  | 2.2425  | 2.7240  | 2.9213  | 1.3487  | 2.9999       | 3.0000     |

$|p_{k,\ell}^{(i)} - p_{k,\ell}^{(i-1)}| < 10^{-8}$ ) for all  $\ell$ . The percolation threshold is determined by identifying the value of  $G$  such that  $\lim_{i \rightarrow \infty} p^{(i)}$  exhibits the first significant jump. The numerical results presented here are obtained using a step size of  $\delta = 0.0001$  for  $G$  and  $L = 40$ . All numerical values are rounded to four decimal places.

### A. Convolutional Coded Poisson Receivers With One Class of Users and One Class of Receivers

In this section, we focus on the single-class system (convolutional coded Poisson Receivers with  $K = J = 1$ ). This section aims to validate the results obtained in Section V.

In Table I, we provide percolation thresholds for the scalar admissible system ( $w = 1$ ) with  $P_{suc}(\rho) = e^{-\rho}$ , i.e., IRSA. These percolation thresholds are evaluated using the density evolution equation in (15). The columns  $w = 2, 3, 4$  present the percolation thresholds  $G_{conv}^*(40, w)$  for  $w = 2, 3, 4$ , respectively, evaluated using the equation in (58). Additionally, we numerically evaluate  $G_s^*$ ,  $G_{conv}^*$ , and  $G_{up}^*$  by their definitions. As for the degree distribution, we set  $\Lambda(x) = x^5$  to reproduce the numerical results presented in [27]. Table II and Table III show the corresponding results for the 2-fold ALOHA and the 3-fold ALOHA, respectively, with  $P_{suc}(\rho)$  being defined in (5)).

From these tables, we observe that  $G_s^* < G_{conv}^* < G_{up}^*$  holds. Also, though the condition of the saturation theorem is not satisfied, the gap between  $G_{conv}^*(L, w)$  and  $G_{conv}^*$  is small. Specifically, the gap ranges from 0.0001 to 0.1. Note that  $G_{conv}^*(L, w)$  and  $G_{up}^*$  in the first two rows of each table match very well with those in Table I of [32] and Table I of [27].

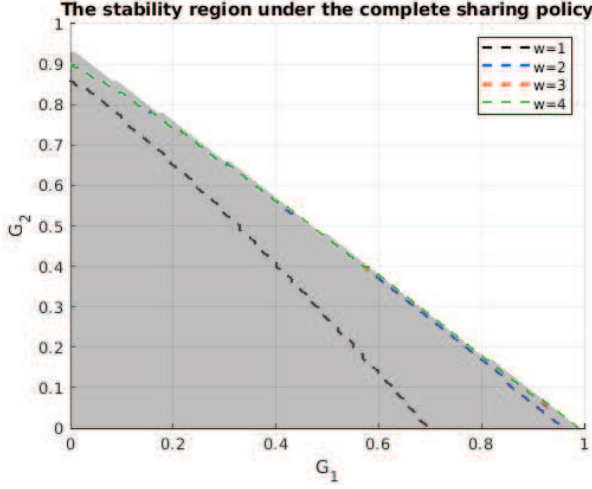


Fig. 2. The boundaries of the stability region of the convolutional IRSA system with two classes of users and two classes of receivers under the complete sharing policy. The gray area presents the outer bound (49) for the complete sharing policy.

### B. IRSA With Two Classes of Users and Two Classes of Receivers

In this section, we consider the IRSA system with two classes of users and two classes of receivers Example 3, with spatial coupling applied. As in Example 3 and [25], we consider the following two packet routing policies:

- 1) Complete sharing: every packet has an equal probability to be routed to the two classes of receivers, i.e.,  $r_{11} = r_{22} = r_{12} = r_{21} = 0.5$ .
- 2) Receiver reservation: class 1 packets are routed to the two classes of receivers with an equal probability, i.e.,  $r_{11} = r_{12} = 0.5$ , and class 2 packets are routed to the class 2 receivers, i.e.,  $r_{21} = 0$ ,  $r_{22} = 1$ .

For our numerical validations, we set the number of iterations of (59) to 10,000. We set the number of stages  $L = 40$  and the window size  $w$  is set from 2 to 4. These numerical results are obtained from a grid search with a step size of  $\delta = 0.01$  for both  $G_1$  and  $G_2$ . For the sake of numerical stability in our computation, we round up  $p_{k,L/2}^{(10000)}(G_1, G_2)$  to 0 if its computed value is smaller than  $10^{-5}$ .

In Figure 2 and Figure 3, we depict the stability region boundaries for two classes of users with two different degree distributions for  $w = 2, 3, 4$ . We choose  $\Lambda_1(x) = x^5$  and  $\Lambda_2(x) = 0.5102x^2 + 0.4898x^4$ , where  $\Lambda_2(x)$  is selected from Table I of [10] to achieve a high percolation threshold of 0.868 in IRSA with a single class of users. The legend  $w = 1$  denotes the conventional coded Poisson receiver (without convolution), reproducing the numerical results in [25]. The colored dashed lines represent the boundary of stability regions for different values of  $w$ . In both figures, the boundaries of the stability regions almost overlap for  $w = 2, 3, 4$ .

Furthermore, the shaded regions in both figures represent the outer bounds of the stability region evaluated in Example 3. Moreover, in both figures, we observe a significant enlargement of the stability region from  $w = 1$  to  $w = 2$ , consistent with the findings of Theorem 5. Therefore, the spatial coupling effect can enlarge the stability

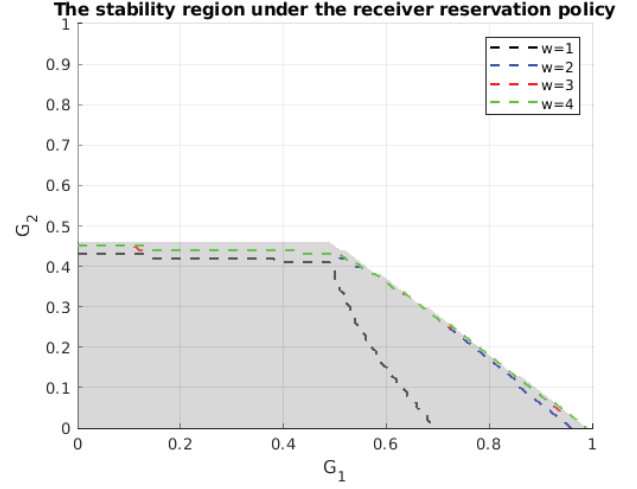


Fig. 3. The boundaries of the stability region of the convolutional IRSA system with two classes of users and two classes of receivers under the receiver reservation policy. The gray area is the intersection of the outer bounds (51) and (52) for the receiver reservation policy.

region. This expansion continues as  $w$  monotonically increases from 2 to 4. Eventually, the boundaries of the expanded stability regions are close to their outer bounds. These empirical results align with the single-class scenario discussed in Section VI-A. For Figure 3, the reservation policy ( $r_{21} = 0$ ) notably constrains the outer bound (52) for  $G_2$ . Thus, the stability region is significantly limited in  $G_2$  compared to that of  $G_1$ .

## VII. CONCLUSION

In this paper, we introduced a probabilistic framework known as convolutional coded Poisson receivers, extending the concept of coded Poisson receivers by drawing inspiration from convolutional LDPC codes. The main contributions of this paper are summarized as follows:

- (i) We established the outer bounds for the stability region of a system of CPRs in Theorem 4, applicable to both CPRs and CCPRs. These outer bounds are extended to encompass multiple traffic classes.
- (ii) We established the saturation theorem (Theorem 7) as a sufficient condition for CCPRs to demonstrate a higher percolation threshold compared to conventional CPRs. We validated the numerical results in Section VI-A.
- (iii) For the single-class user and receiver scenario of CCPRs, we employed the potential function to characterize three important thresholds, namely  $G_s^* < G_{conv}^* < G_{up}^*$ , as presented in Theorem 9.
- (iv) Our numerical results showed that the stability region of CCPRs might approach its outer bounds under finite iterations of the density evolution equations.

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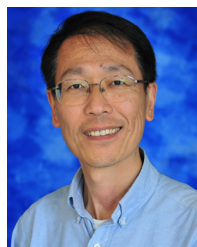
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